Engineering Capacity Planning & Resource Allocation Platform

Role: Technical Program Manager — independently designed and built the tool

Focus: Engineering capacity modeling, resource allocation, portfolio planning, and delivery governance

Executive summary

I independently designed and built an interactive engineering capacity planning tool to help engineering and program leaders compare proposed project demand with realistically available team capacity. The tool combined individual allocation percentages, available working days, and other capacity adjustments; translated engineering effort estimates into FTE-equivalent demand; and highlighted teams whose planned allocations exceeded available capacity.

The system supported more informed discussions about scope, staffing, sequencing, and roadmap commitments. It also connected the planning view to project context and execution-health signals. No quantitative improvement in planning accuracy or decision speed is asserted because those measures were not supplied.

Business and operational problem

Planning decisions were distributed across project estimates, team staffing, resource allocations, and delivery commitments. Teams needed a shared way to understand whether proposed commitments fit within available engineering capacity, especially where individuals were split across projects and teams.

Core questions: How much capacity is available? How much is already allocated? Where are the gaps? Which commitments need a staffing, scope, or sequencing decision?

Architecture and information flow

Individual allocation inputs + working days + adjustments

↓ Effective team capacity

↓ Project / epic effort estimates → normalized demand

↓ Capacity vs. demand comparison

↓ Utilization and gap indicators

↓ Interactive dashboard → leadership decisions

Calculation model

Confirmed design principles supplied by the builder:

- Start with the number of working days in each month of the planning period.
- Subtract regional holidays and individual planned time off.
- Apply a **15% efficiency shrink**, leaving **85%** of net available days as effective delivery capacity.

- Account for the percentage of each engineer allocated to the team or workstream.
- Convert estimated engineering effort into FTE-equivalent demand, using a consistent time period and units.
- Calculate utilization as allocated effort divided by total available person-days.

Capacity formulas (generalized):

- Net available days per person = monthly working days − regional holidays − planned time off
- Effective days per person = max(0, net available days) × 0.85
- Team-allocated person-days = effective days per person × allocation fraction
- Team effective capacity = sum of team-allocated person-days across people
- Utilization % = allocated project effort (person-days) / team effective capacity (person-days) × 100
- Capacity gap = team effective capacity − allocated project effort

If effort is represented as FTE-equivalent demand, normalize it into person-days for the same period before comparing it with capacity. The precise FTE conversion denominator is not documented in the supplied materials, so it is not asserted as a production rule.

Implementation boundary: The source and builder confirm the inputs and 15% shrink, but not the precise handling of overlapping allocations, different employment categories, or whether the original tool applied allocation before or after the shrink. Multiplication is equivalent under the simple model above; more complex business rules may change that.

Illustrative calculation walkthrough — fictional values

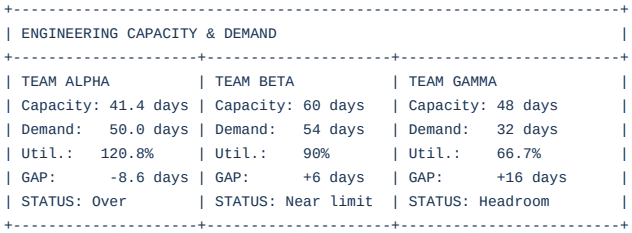
Assume a team has three engineers, each with **22 working days**, **1 regional holiday**, and **2 days of planned leave** in the month. Their allocations to this team are **100%**, **80%**, and **50%**.

- Net days per engineer: 22 − 1 − 2 = 19 days
- Effective days after 15% shrink: 19 × 0.85 = 16.15 days
- Total allocated team capacity: 16.15 × (1.00 + 0.80 + 0.50) = 37.145 person-days
- Proposed engineering effort: 45 person-days
- Utilization: 45 / 37.145 × 100 ≈ 121.1%
- Capacity shortfall: 45 − 37.145 = 7.855 person-days

This is a teaching example, **not** actual internal staffing, allocations, or program results.

Example dashboard wireframe (fictional)

Illustrative dashboard



Decision-support scenario

Illustrative scenario: Team Alpha has 41.4 available person-days but 50 person-days of planned work. Rather than accepting the roadmap at face value, a program leader can use the dashboard to surface the 8.6-day shortfall and facilitate a decision: move a lower-priority item, adjust timing, reassign eligible work, or revisit scope and estimates. The tool makes the constraint explicit; leadership retains the decision.

Design and implementation decisions

- **Normalize demand and supply:** Compare effort and capacity in consistent units rather than mixing FTE, person-days, and story points.
- **Account for partial allocations:** Reflect that an individual may be available for only a fraction of a planning period.
- **Expose assumptions:** Make capacity adjustments visible rather than implying full nominal staffing is always usable.
- **Support team-level exploration:** Filter and sort by project or team and show utilization status.
- **Connect planning to execution:** Surface active projects and execution-health indicators alongside capacity.
- **Keep the tool lightweight:** The documented implementation used a self-contained browser interface with structured project data; no always-on backend was required for display.

My contribution

I personally built the solution from scratch, including its dashboard, calculation logic, and interactive planning experience. This was hands-on tool design and implementation, not only requirements gathering or coordination of another engineering team's development.

As a TPM, I used systems thinking to translate planning friction into a practical decision-support mechanism connecting engineering demand, staffing availability, and leadership trade-offs.

Why this experience transfers to a platform TPM role

The underlying problem is common to complex engineering organizations: demand exceeds finite capacity, dependencies cross team boundaries, and roadmap commitments require explicit trade-offs. This case demonstrates an ability to build mechanisms that improve decision quality while coordinating planning and delivery at scale.

Evidence boundaries and follow-up questions

- Confirm any additional capacity adjustments beyond regional holidays, time off, and the 15% efficiency shrink.
- Confirm how staffing changes, time off, contractors, and double-booking are handled.
- Confirm conversion rules for effort estimates and whether project demand was recorded in person-months or another unit before normalization.
- Confirm thresholds used for utilization indicators.
- Add verified adoption or efficiency metrics only if supported by evidence.